A subcritical phase for the Cox Boolean model on Poisson lines

Working note: a multi-scale proof attempt, not yet independently checked

Christian Hirsch* Benedikt Jahnel†

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Abstract

We consider the Boolean model with unit disks centred at the points of a Cox process of intensity λ per unit length on an isotropic Poisson line process of intensity μ in the plane. This note contains a complete argument for the existence of a subcritical phase, $\lambda_c(\mu) > 0$ for every $\mu > 0$. The argument is a multi-scale recursion with scale ratio $M = \alpha^4$ and 25 well-separated sub-crossings per step. Its ingredients are: the domination of the occupied boxes by the boxes close to at least j lines and an independent Bernoulli field; an exact conditional independence of the sub-crossings given the lines shared by two or more of the regions; the observation that a shared line is seen by its first two regions as a line of fixed direction with independent uniform offsets, so that by the Mecke formula it costs only a logarithmic factor there; and an induction hypothesis for crossings in the presence of s arbitrarily placed additional lines, with a bound that deteriorates only by a small power per line. The note is a proof attempt: every step is written out and has been checked by its authors' tools, but it has not been checked independently, and the result should not be regarded as established before that.

1 Result

Let $\mathcal{L}$ be a Poisson line process in $\mathbb{R}^2$ with intensity measure $\nu(\mathrm{d}\theta\,\mathrm{d}p) = (\mu/\pi)\,\mathrm{d}\theta\,\mathrm{d}p$, where a line $\{y\colon \langle y, n_\theta\rangle = p\}$ is parametrised by the direction $\theta \in [0,\pi)$ of its unit normal n_θ and by $p \in \mathbb{R}$; the lines meeting a convex set of perimeter P form a Poisson variable with mean $\mu P/\pi$. Conditionally on $\mathcal{L}$, let X be a Poisson process of intensity λ per unit length on every line of $\mathcal{L}$, and let $\Xi := \bigcup_{c\in X} B(c,1)$. A *chain* is a finite sequence of points of X with consecutive points at distance less than 2.

Theorem 1.1 (Subcritical phase)**.** *For every $\mu > 0$ there are $\lambda_0 = \lambda_0(\mu) > 0$ and $\alpha_0 = \alpha_0(\mu)$ such that for all $\lambda \le \lambda_0$, all $n \ge 0$ and all $x \in \mathbb{R}^2$, with $\alpha_n := \alpha_0^{5^n}$, the probability that there is a chain from a point of $B(x, \alpha_n - 1)$ to a point outside $B(x, 8\alpha_n + 1)$ is at most α_n^{-7}. In particular, for $\lambda \le \lambda_0$ almost surely all connected components of Ξ are bounded, so that $\lambda_c(\mu) \ge \lambda_0(\mu) > 0$.*

*Institut für Mathematische Stochastik, Georg-August-Universität Göttingen, Goldschmidtstraße 7, 37077 Göttingen, Germany. christian.hirsch@uni-goettingen.de. ORCID 0000-0003-4136-3740.

†Institut für Mathematische Stochastik, Technische Universität Braunschweig, Universitätsplatz 2, 38106 Braunschweig, Germany, and Weierstrass Institute for Applied Analysis and Stochastics, Mohrenstraße 39, 10117 Berlin, Germany. benedikt.jahnel@tu-braunschweig.de. ORCID 0000-0002-4212-0065.

The second assertion follows from the first: an unbounded component meeting $B(0,R)$ contains, for every n with $\alpha_n \geq R+2$, a chain from $B(0,\alpha_n-1)$ to the complement of $B(0,8\alpha_n+1)$, so its probability is at most $\inf_n \alpha_n^{-7} = 0$.

Parameters. Throughout,

$$\begin{gathered}\kappa := 4, \qquad a := 25, \qquad \gamma_0 := 7, \qquad \gamma_* := 0.11, \qquad \sigma := 0.005,\\ \bar\sigma := 0.12, \qquad k_c := 1378, \qquad \xi := 0.1, \qquad p := \tfrac{1}{48}\,,\end{gathered} \tag{1.1}$$

and

$$\gamma(k) := \gamma_0 - \sigma k \quad (0 \leq k \leq k_c), \qquad \gamma(k) := 0 \quad (k > k_c)\,, \tag{1.2}$$

so that γ is non-increasing, $\gamma(k) \geq \gamma(k_c) = \gamma_*$ for $k \leq k_c$, and $\sigma + \gamma_* \leq \bar\sigma$. The following inequalities hold:

$$(a-\kappa-1)(\gamma_0-\kappa) \geq (1+\kappa)\gamma_0 + \tfrac{\kappa}{2} + \xi\,, \qquad \gamma_0 \geq \tfrac{3}{2}\kappa + \xi\,, \tag{1.3}$$

$$(a-1-\kappa)\gamma_* \geq \tfrac{\kappa}{2} + \xi\,, \qquad \bar\sigma(a-2) \leq \kappa - 1 - 0.2\,. \tag{1.4}$$

Indeed $60 \geq 37.1$, $7 \geq 6.1$, $2.2 \geq 2.1$ and $2.76 \leq 2.8$.

Lemma 1.2 (Profile)**.** *For all integers $k \geq 0$ and $b \geq 0$, $\gamma(k) - \gamma(k+b) \leq \bar\sigma b$.*

Proof. Let $b \geq 1$. If $k + b \leq k_c$ the difference is σb, and if $k > k_c$ it is 0. If $k \leq k_c < k + b$, it equals $\gamma(k) = \gamma_* + \sigma(k_c - k) \leq \gamma_* + \sigma(b-1) \leq (\gamma_* + \sigma)b \leq \bar\sigma b$. □

Plan. Section 2 defines the lattice model, the crossing events with additional lines and the base case. Section 3 contains the geometry: the inclusion of a crossing in a sub-crossings, the number of regions met by a line, and coordinates for the lines meeting two regions. Section 4 proves the decoupling of a sub-crossings. Section 5 contains the induction over scales and the proof of Theorem 1.1. Section 6 lists the points at which an error would be most likely.

2 Setting

Counts and crossing events. For a site $z \in \mathbb{Z}^2$, a radius $r \in [1,2]$ and a finite set Λ of lines put

$$N_r^\Lambda(z) := \#\{\ell \in \mathcal{L}\colon \operatorname{dist}(z,\ell) \leq r\} + \#\{\ell \in \Lambda\colon \operatorname{dist}(z,\ell) \leq r\}\,,$$

and $N_r := N_r^\varnothing$, a Poisson variable with mean $2\mu r \leq 4\mu$. Let $U = (U_z)_{z\in\mathbb{Z}^2}$ be independent uniform variables on $[0,1]$, independent of $\mathcal{L}$, and $\omega := \{z\colon U_z \leq p\}$. A *path* is a finite sequence of sites with consecutive sites at sup-distance at most 2; a step has Euclidean length at most $2\sqrt{2} < 3$.

Definition 2.1 (Crossing events)**.** For $x \in \mathbb{R}^2$, $\rho \geq 5$, $j \in \mathbb{Z}$, $r \in [1,2]$ and a finite set Λ of lines, $G_j^{r,\Lambda}(x,\rho)$ is the event that there is a path $z_1,\dots,z_q$ of sites in $\{z\colon N_r^\Lambda(z) \geq j\} \cup \omega$ with $|z_1 - x| \leq \rho$, $|z_q - x| > 8\rho$ and $|z_i - x| \leq 8\rho$ for $i < q$. For $s \geq 0$ put

$$F_j^r(s;\rho) := \sup_{x\in\mathbb{R}^2} \sup_{|\Lambda|\leq s} \mathbb{P}\big(G_j^{r,\Lambda}(x,\rho)\big)\,.$$

The events are increasing in r, in Λ and in $\mathcal{L}$, and decreasing in j; $F_j^r(s;\rho)$ is non-decreasing in r and s. All sites of a path as above lie in $A(x,\rho) := \{z\colon |z-x| \le 8\rho+3\}$, and a line within distance $r \le 2$ of such a site meets $B(x,9\rho)$ because $8\rho+5 \le 9\rho$. Hence $G_j^{r,\Lambda}(x,\rho)$ is determined by $(U_z)_{z\in A(x,\rho)}$ and by the lines of $\mathcal{L}\cup\Lambda$ meeting $B(x,9\rho)$. Since $N_r^\Lambda \le N_r + |\Lambda|$,

$$F_j^r(s;\rho) \le F_{j-s}^r(0;\rho)\,. \tag{2.1}$$

Lemma 2.2 (Domination)**.** *Let $j \ge 1$ and $\sqrt{2}\,\lambda(j-1) \le p$. Then for all $x \in \mathbb{R}^2$ and $\rho \ge 5$ the probability that there is a chain from a point of $B(x,\rho-1)$ to a point outside $B(x,8\rho+1)$ is at most $F_j^1(0;\rho)$.*

Proof. Let $Q(z) := z + [-1/2,1/2)^2$ and let $N(z) \le N_1(z)$ be the number of lines of $\mathcal{L}$ meeting $Q(z)$. Conditionally on $\mathcal{L}$, the events $\{X \cap Q(z) \ne \varnothing\}$, $z \in \mathbb{Z}^2$, are independent with probabilities $q_z \le \min\{1, \sqrt{2}\lambda N(z)\}$, since a chord of a unit square has length at most $\sqrt{2}$. Let (V_z) be independent uniform variables independent of $(\mathcal{L}, X)$, and put $U_z := q_z V_z$ if $X \cap Q(z) \ne \varnothing$ and $U_z := q_z + (1-q_z)V_z$ otherwise. Conditionally on $\mathcal{L}$ the U_z are independent and uniform, so U is independent of $\mathcal{L}$ and has the required law. If $X \cap Q(z) \ne \varnothing$ and $N_1(z) \le j-1$, then $U_z \le q_z \le \sqrt{2}\lambda(j-1) \le p$, so $z \in \omega$. Hence every site whose box contains a point of X lies in $\{N_1 \ge j\} \cup \omega$. If $c_1, \dots, c_q$ is a chain as in the statement and $c_i \in Q(z_i)$, then $|z_i - c_i| < 1$, consecutive sites z_i are at sup-distance less than 3, hence at most 2, $|z_1 - x| \le \rho$ and $|z_q - x| > 8\rho$; stopping at the first site at distance more than 8ρ from x gives a path as in Definition 2.1. □

Lemma 2.3 (Base case)**.** *For every $\mu > 0$ and $\alpha_0 \ge 40$ there is $j_0 = j_0(\mu,\alpha_0)$ such that $F_{j_0}^2(0;\alpha_0) \le \alpha_0^{-\gamma_0}$. Consequently, with $j := j_0 + k_c$, $F_j^2(s;\alpha_0) \le \alpha_0^{-\gamma(s)}$ for all $s \ge 0$.*

Proof. Fix x. On $G_{j_0}^{2,\varnothing}(x,\alpha_0)$ either some site of $A(x,\alpha_0)$ has $N_2 \ge j_0$, or there is a path as in Definition 2.1 in ω. The set $A(x,\alpha_0)$ has at most $\pi(8\alpha_0+4)^2 \le 300\alpha_0^2$ sites, so the first alternative has probability at most $300\alpha_0^2\,\mathbb{P}(\mathrm{Poi}(4\mu) \ge j_0) \le \alpha_0^{-7}/2$ for j_0 large. In the second alternative the path may be taken self-avoiding; it starts at one of at most $4\alpha_0^2$ sites and has at least $\lceil 2.4\alpha_0 \rceil$ sites, because it covers a distance of more than $7\alpha_0$ in steps of length at most $2\sqrt{2}$. There are at most 24^{q-1} self-avoiding paths with q sites from a given start, so the probability is at most $4\alpha_0^2 \sum_{q \ge 2.4\alpha_0} 24^{q-1} 48^{-q} \le 4\alpha_0^2\, 2^{-2.4\alpha_0} \le \alpha_0^{-7}/2$ for $\alpha_0 \ge 40$. The consequence follows from (2.1) for $s \le k_c$ and is trivial for $s > k_c$. □

3 Geometry

Let $M \ge \max\{100, 6a\}$ and $\alpha \ge 5$. For $i = 1, \dots, a$ let $\beta_i := 1 + 7(i-1)/(a-1)$, so that $\beta_1 = 1$ and $\beta_a = 8$, and let K_i be a set of equally spaced points on the circle of radius $M\beta_i$ around the origin with arc spacing in $[0.48, 0.5]$; then every point of this circle is within distance $1/4$ of K_i and $|K_i| \le 13M\beta_i \le 104M$. A *tuple* is an element $T = (k_1, \dots, k_a)$ of $K_1 \times \cdots \times K_a$.

Lemma 3.1 (Inclusion)**.** *For all $x \in \mathbb{R}^2$, j, r and Λ,*

$$G_j^{r,\Lambda}(x, M\alpha) \subseteq \bigcup_T \bigcap_{i=1}^{a} G_j^{r,\Lambda}(x + \alpha k_i, \alpha)\,, \tag{3.1}$$

and $|(x+\alpha k_i) - (x+\alpha k_{i'})| \ge D := 7M\alpha/(a-1) \ge 36\alpha$ for $i \ne i'$.

Proof. Let $z_1, \dots, z_q$ be a path as in the definition of the left side, and let $i < a$. Let m be the largest index with $|z_m - x| \le M\beta_i\alpha$; then $m < q$ and $|z_m - x| > M\beta_i\alpha - 3$, so there is $k_i \in K_i$ with $|z_m - x_i| \le 3 + \alpha/4 \le \alpha$, where $x_i := x + \alpha k_i$. Since $|z_q - x_i| > 8M\alpha - M\beta_i\alpha \ge D \ge 8\alpha$, the sequence $z_m, z_{m+1}, \dots$ stopped at its first site at distance more than 8α from x_i realises $G_j^{r,\Lambda}(x_i, \alpha)$. For $i = a$ we have $8M\alpha < |z_q - x| \le 8M\alpha + 3$, so there is $k_a \in K_a$ with $|z_q - x_a| \le \alpha$, and the reversed sequence $z_q, z_{q-1}, \dots$, which ends at z_1 with $|z_1 - x_a| \ge 7M\alpha > 8\alpha$, stopped at its first site at distance more than 8α from x_a, realises $G_j^{r,\Lambda}(x_a, \alpha)$. The distance between points on different circles is at least the difference of the radii. □

For a tuple T and $x \in \mathbb{R}^2$ write $x_i := x + \alpha k_i$ and $R_i := B(x_i, 9\alpha)$; the *regions* R_i are pairwise disjoint. We say that a line ℓ is *near-tangent* to the ith circle if its distance δ from x satisfies $|\delta - M\beta_i\alpha| < w\alpha$ with $w := 7M/(2(a-1)) \ge 18$. Since the radii differ by $2w\alpha$, a line is near-tangent to at most one circle.

Lemma 3.2 (Regions met by a line)**.** *Let ℓ be a line and $1 \le i \le a$. The number of $k \in K_i$ with $\ell \cap B(x + \alpha k, 9\alpha) \ne \varnothing$ is at most $c_1\sqrt{M}$ with $c_1 := 80$, and at most $c_0 := 163\sqrt{a}$ if ℓ is not near-tangent to the ith circle.*

Proof. Scale by $1/\alpha$ and translate x to the origin; let $R := M\beta_i \in [M, 8M]$ and δ the distance of the line from the origin. With φ the angle of a point of the circle measured from the normal direction of the line, the point is within distance 9 of the line if and only if $\cos\varphi \in I := [(\delta-9)/R, (\delta+9)/R]$. The set of $\varphi \in [0, \pi]$ with $\cos\varphi \in I$ has measure $\int_{I \cap [-1,1]} (1-u^2)^{-1/2}\,\mathrm{d}u$. In general this is at most $\arccos(1 - 18/R) = 2\arcsin\sqrt{9/R} \le 6.6/\sqrt{R}$, because the integrand is largest at the ends of $[-1, 1]$ and $\arcsin s \le 1.1s$ for $s \le 0.3$; by symmetry the points lie on at most two arcs of total length at most $13.2\sqrt{R}$, which contain at most $13.2\sqrt{R}/0.48 + 2 \le 28\sqrt{R} \le 80\sqrt{M}$ points of K_i. If ℓ is not near-tangent and $\delta \ge R + w$, no point of the circle is within distance 9 of the line. If $\delta \le R - w$, then on I we have $u \le 1 - (w-9)/R \le 1 - w/(2R)$ and $u \ge -9/R$, hence $1 - u^2 \ge w/(2R)$, and the measure is at most $(18/R)\sqrt{2R/w}$. The two arcs have total length at most $36\sqrt{2R/w} \le 36\sqrt{32(a-1)/7} \le 77\sqrt{a}$ and contain at most $77\sqrt{a}/0.48 + 2 \le 163\sqrt{a}$ points of K_i. □

For $y \in \mathbb{R}^2$, a unit vector e with $e^\perp$ its rotation by $\pi/2$, and $u \in \mathbb{R}$ let

$$\ell_y^e(u) := \{y + ue^\perp + te \colon t \in \mathbb{R}\},$$

the line of direction e at signed distance u from y.

Lemma 3.3 (Lines meeting two regions)**.** *Let $y_1, y_2 \in \mathbb{R}^2$ with $D' := |y_1 - y_2| \ge D \ge 36\alpha$, $e := (y_2 - y_1)/D'$, and let $\mathcal{W}$ be the set of lines meeting both $B(y_1, 9\alpha)$ and $B(y_2, 9\alpha)$.*

1. *There is an injective map $\ell \mapsto (u_1, u_2)$ from $\mathcal{W}$ to $[-9\alpha, 9\alpha]^2$ under which the image of $\nu|_{\mathcal{W}}$ has a density with respect to Lebesgue measure bounded by $1.25\mu/(\pi D)$.*
2. *For $i \in \{1, 2\}$, every $\ell \in \mathcal{W}$, every $\varrho \le 2$ and every site z with $|z - y_i| \le 8\alpha + 3$: if $\operatorname{dist}(z, \ell) \le \varrho$, then $\operatorname{dist}(z, \ell_{y_i}^e(u_i)) \le \varrho + \eta$ with $\eta := 335\alpha^2/D$.*

Proof. Write the unit normal of a line as $n = \cos\varphi\, e^\perp + \sin\varphi\, e$ with $\varphi \in (-\pi/2, \pi/2]$, and let $u_i := p - \langle y_i, n\rangle$ be the signed distance of the line from y_i. The map $(\theta, p) \mapsto (\varphi, u_1)$ is a rotation in the first and a translation in the second coordinate, so ν has density μ/π in (φ, u_1). A line belongs to $\mathcal{W}$ if and only if $|u_1|, |u_2| \le 9\alpha$, and $u_1 - u_2 = \langle y_2 - y_1, n\rangle = D'\sin\varphi$. Hence on $\mathcal{W}$, $|\sin\varphi| \le 18\alpha/D' \le 1/2$, the map $(\varphi, u_1) \mapsto (u_1, u_2)$ is injective with Jacobian $D'\cos\varphi \ge 0.8D'$, and

(i) follows. For (ii) let $d \coloneqq \cos\varphi\, e - \sin\varphi\, e^{\perp}$ be the direction of ℓ, so that $\ell = \{y_i + u_i n + td\colon t \in \mathbb{R}\}$. Let g be the point of ℓ nearest to z; then $|g - y_i| \le 8\alpha + 5 \le 9\alpha$, so $g = y_i + u_i n + td$ with $|u_i|, |t| \le 9\alpha$. The point $g' \coloneqq y_i + u_i e^{\perp} + te$ lies on $\ell^{e}_{y_i}(u_i)$ and $|g - g'| \le (|u_i| + |t|)\cdot 2\sin(|\varphi|/2) \le 18\alpha \cdot 1.035|\sin\varphi| \le 335\alpha^2/D'$, because $2\sin(\varphi/2) = \sin\varphi/\cos(\varphi/2)$ and $|\varphi| \le \pi/6$. □

4 Decoupling of the sub-crossings

Proposition 4.1 (Decoupling)**.** *Let $\alpha \ge 5$, let $x_1, \dots, x_a \in \mathbb{R}^2$ have pairwise distances at least $D \ge 36\alpha$, and let $\Lambda_1, \dots, \Lambda_a$ be finite sets of lines with $s_i \coloneqq |\Lambda_i|$. Let $r \ge 1$, $\eta \coloneqq 335\alpha^2/D$ and $\eta' > 0$ be such that $r'' \coloneqq r + \eta + \eta' \le 2$, and put $m \coloneqq 2\mu\eta'/\pi$, $q \coloneqq 129\mu\alpha^2/D$. Assume $(a-1)m \le 1/2$, let $t \ge 1$ be an integer and $\Theta(k) \coloneqq F_j^{r''}(k;\alpha)$. Then*

$$\mathbb{P}\Big(\bigcap_{i=1}^{a} G_j^{r,\Lambda_i}(x_i,\alpha)\Big) \le \sum_{h} \prod_{k<l} \frac{(q\,t^2/m^2)^{h_{kl}}}{h_{kl}!} \prod_{i=1}^{a} \big[\Theta\big(s_i + b_i(h)\big) + 2^{-t/2}\big]\,, \tag{4.1}$$

where the sum is over all families $h = (h_{kl})_{1\le k<l\le a}$ of non-negative integers and $b_i(h) \coloneqq \sum_{k<l<i} h_{kl}$.

Proof. Write $R_i \coloneqq B(x_i, 9\alpha)$, $A_i \coloneqq A(x_i,\alpha)$ and $G_i \coloneqq G_j^{r,\Lambda_i}(x_i,\alpha)$. The regions R_i are pairwise disjoint and so are the sets A_i.

Step 1: conditioning on the shared lines. Let $\mathcal{S}$ be the set of lines of $\mathcal{L}$ meeting at least two of the regions, and for $\ell \in \mathcal{S}$ let $k(\ell) < l(\ell)$ be the two smallest indices of regions met by ℓ. The sets $\mathcal{S}_{kl} \coloneqq \{\ell \in \mathcal{S}\colon (k(\ell), l(\ell)) = (k,l)\}$, $k<l$, and the sets $\mathcal{L}_i$ of lines of $\mathcal{L}$ meeting R_i and no other region, $1 \le i \le a$, are restrictions of $\mathcal{L}$ to pairwise disjoint sets of lines, hence independent Poisson processes; the intensity measure of $\mathcal{S}_{kl}$ is at most ν restricted to the set $\mathcal{W}_{kl}$ of lines meeting R_k and R_l. The event G_i is determined by $\mathcal{L}_i$, by the lines of $\mathcal{S}$ meeting R_i and by $(U_z)_{z\in A_i}$. Therefore the events $G_1, \dots, G_a$ are conditionally independent given $\mathcal{S}$, and with $\pi_i(\mathcal{S}) \coloneqq \mathbb{P}(G_i \,|\, \mathcal{S})$,

$$\mathbb{P}\Big(\bigcap_i G_i\Big) = \mathbb{E}\Big[\prod_{i=1}^{a} \pi_i(\mathcal{S})\Big]\,. \tag{4.2}$$

Step 2: a pointwise bound. For a region i, a radius ϱ, a finite set Λ' of lines, and unit vectors $e_1, \dots, e_v$ and reals $u_1, \dots, u_v$ put

$$\Phi_i^{\varrho}(u_1, \dots, u_v; \Lambda') \coloneqq \mathbb{P}\Big(G_j^{\varrho,\, \Lambda_i \cup \Lambda' \cup \{\ell^{e_1}_{x_i}(u_1), \dots, \ell^{e_v}_{x_i}(u_v)\}}(x_i,\alpha)\Big)\,,$$

the dependence on the directions being suppressed. Fix a realisation of $\mathcal{S}$. For $\ell \in \mathcal{S}_{kl}$ let $(u_k(\ell), u_l(\ell))$ be its coordinates from Lemma 3.3 applied to $y_1 = x_k$, $y_2 = x_l$, and e_{kl} the corresponding direction. In region i we call $\ell \in \mathcal{S}$ *straightened* if $i \in \{k(\ell), l(\ell)\}$, and *free* if ℓ meets R_i and $i > l(\ell)$. Since $\mathcal{L}_i \subseteq \mathcal{L}$ and the events are increasing in the set of lines, $\pi_i(\mathcal{S})$ is at most the probability of G_i computed with an independent copy of $\mathcal{L}$ in place of $\mathcal{L}_i$ and with the lines of $\mathcal{S}$ meeting R_i added to Λ_i. By Lemma 3.3 (ii) and monotonicity in the radius, replacing each straightened line ℓ by $\ell^{e_{kl}}_{x_i}(u_i(\ell))$ and the radius r by $r+\eta$ does not decrease this probability. Hence

$$\pi_i(\mathcal{S}) \le \Phi_i^{r+\eta}\big((u_i(\ell))_{\ell \text{ straightened in } i};\, \{\ell\colon \ell \text{ free in } i\}\big)\,. \tag{4.3}$$

The number of lines that are free in i is at most $b_i(h)$, where $h_{kl} \coloneqq |\mathcal{S}_{kl}|$, and the number of straightened lines in i is $v_i(h) \coloneqq \sum_{l>i} h_{il} + \sum_{k<i} h_{ki}$.

Step 3: integration. Put

$$\bar\Phi_i(v,b) := \sup\ \frac{1}{(18\alpha)^v}\int_{[-9\alpha,9\alpha]^v} \Phi_i^{r+\eta}(u_1,\dots,u_v;\Lambda')\,\mathrm{d}u\,,$$

the supremum being over all directions $e_1,\dots,e_v$ among the at most $a-1$ vectors e_{il}, e_{ki}, and over all sets Λ' of at most b lines. We claim that

$$\mathbb{E}\Big[\prod_{i=1}^a \pi_i(\mathcal{S})\Big] \le \sum_h \prod_{k<l}\frac{q^{h_{kl}}}{h_{kl}!}\ \prod_{i=1}^a \bar\Phi_i(v_i(h),b_i(h))\,. \tag{4.4}$$

By the independence of the Poisson processes $\mathcal{S}_{kl}$, the bound on their intensity measures, (4.3) and Lemma 3.3 (i), the left side is at most

$$\sum_h \prod_{k<l}\frac{1}{h_{kl}!}\Big(\frac{1.25\mu}{\pi D}\Big)^{h_{kl}} \int \prod_{i=1}^a \Phi_i^{r+\eta}((u_i(\ell))_{\ell\text{ str. in }i};\,\{\ell\text{ free in }i\}) \prod_\ell \mathrm{d}u_{k(\ell)}(\ell)\,\mathrm{d}u_{l(\ell)}(\ell)\,,$$

where for given h the integral is over the coordinates of h_{kl} lines in each class (k,l), each pair of coordinates ranging over the image of $\mathcal{W}_{kl}$ in $[-9\alpha,9\alpha]^2$, and the set of lines that are free in i is a function of the coordinates of the lines of the classes (k,l) with $l<i$. Whenever a coordinate of a line is integrated below, the range of integration is enlarged to $[-9\alpha,9\alpha]$; this is legitimate because the integrand is non-negative and because, at that moment, the remaining factors depend on the line only through its other coordinate. We integrate the variables in the order of decreasing region index. The variables $u_a(\ell)$ belong to lines of the classes (k,a). They occur in the factor of region a only: in region k such a line enters through $u_k(\ell)$, and it is free in no region. The lines that are free in region a depend on other variables. Integrating over the $v_a(h)$ variables $u_a(\ell)$ with all other variables fixed therefore bounds the factor of region a by $(18\alpha)^{v_a(h)}\bar\Phi_a(v_a(h),b_a(h))$. Suppose the factors of the regions $i+1,\dots,a$ have been bounded in this way. The variables $u_i(\ell)$ belong to lines of the classes (k,i) and (i,l). Such a line is free only in regions of index larger than i, whose factors have already been replaced by constants, and it enters the factor of its other region through its other coordinate. So the variables $u_i(\ell)$ occur in the factor of region i only, and the lines that are free in region i belong to classes (k,l) with $l<i$ and do not depend on them. Integrating gives the factor $(18\alpha)^{v_i(h)}\bar\Phi_i(v_i(h),b_i(h))$. Since $\sum_i v_i(h) = 2\sum_{k<l}h_{kl}$ and $1.25\mu(18\alpha)^2/(\pi D)\le q$, this proves (4.4).

Step 4: the Mecke comparison. We show that for all $v,b\ge 0$,

$$\bar\Phi_i(v,b) \le \Big(\frac{t}{m}\Big)^v\big[\Theta(s_i+b)+2^{-t/2}\big]\,. \tag{4.5}$$

Together with (4.2), (4.4) and $\sum_i v_i(h) = 2\sum_{k<l}h_{kl}$ this gives (4.1). Fix directions $e_1,\dots,e_v$ and Λ' with $|\Lambda'|\le b$, and put $\psi := \eta'/(18\alpha)$. For a unit vector e let $\mathcal{W}(e)$ be the set of lines whose normal is $\cos\varphi\, e^\perp + \sin\varphi\, e$ with $|\varphi|\le\psi$ and whose signed distance u from x_i satisfies $|u|\le 9\alpha$; in the coordinates (φ,u) the measure ν has density μ/π on $\mathcal{W}(e)$, so $\nu(\mathcal{W}(e)) = (\mu/\pi)\cdot 2\psi\cdot 18\alpha = m$. As in the proof of Lemma 3.3 (ii), for a line $\ell\in\mathcal{W}(e)$ with coordinates (φ,u) and a site $z\in A_i$, $\mathrm{dist}(z,\ell^e_{x_i}(u))\le\varrho\le 2$ implies $\mathrm{dist}(z,\ell)\le\varrho+18\alpha|\varphi|\le\varrho+\eta'$. Hence for every choice of $\varphi_1,\dots,\varphi_v\in[-\psi,\psi]$ the value $\Phi_i^{r+\eta}(u_1,\dots,u_v;\Lambda')$ is at most the probability of $G_j^{r'',\Lambda_i\cup\Lambda'\cup\{\ell_1,\dots,\ell_v\}}(x_i,\alpha)$, where $\ell_c\in\mathcal{W}(e_c)$ has coordinates (φ_c,u_c). Averaging over $\varphi_1,\dots,\varphi_v$ and writing $g(\mathcal{L})$ for the indicator of $G_j^{r'',\Lambda_i\cup\Lambda'}(x_i,\alpha)$ as a function of the line process, we obtain

from the multivariate Mecke formula [LP17, Theorem 4.4]

$$\frac{1}{(18\alpha)^v}\int \Phi_i^{r+\eta}(u;\Lambda')\,\mathrm{d}u \le \frac{1}{m^v}\int \mathbb{E}[g(\mathcal{L}\cup\{\ell_1,\dots,\ell_v\})]\prod_{c=1}^{v}\mathbf{1}\{\ell_c\in\mathcal{W}(e_c)\}\,\nu^{\otimes v}(\mathrm{d}\ell)$$
$$= \frac{1}{m^v}\,\mathbb{E}\Big[\sum_{\ell_1,\dots,\ell_v\in\mathcal{L}}^{\neq}\prod_{c=1}^{v}\mathbf{1}\{\ell_c\in\mathcal{W}(e_c)\}\;g(\mathcal{L})\Big] \le \frac{1}{m^v}\,\mathbb{E}\big[(N^*)^v\,g(\mathcal{L})\big]\,,$$

where the sum is over tuples of pairwise distinct lines and N^* is the number of lines of $\mathcal{L}$ in the union of the sets $\mathcal{W}(e)$ over the at most $a-1$ admissible directions, a Poisson variable with mean $m^*\le(a-1)m\le 1/2$. Since $\mathbb{E}[g(\mathcal{L})]\le\Theta(s_i+b)$,

$$\mathbb{E}\big[(N^*)^v g(\mathcal{L})\big] \le t^v\,\Theta(s_i+b)+\mathbb{E}\big[(N^*)^v\mathbf{1}\{N^*>t\}\big]\,.$$

If $v\le t/2$, then $n^v\le 2^v\,n(n-1)\cdots(n-v+1)$ for $n>t$, and

$$\mathbb{E}\big[(N^*)^v\mathbf{1}\{N^*>t\}\big]\le 2^v(m^*)^v\,\mathbb{P}(N^*>t-v)\le 2^v(m^*)^v(m^*)^{t/2}\le t^v\Big(\frac{2(a-1)m}{t}\Big)^v 2^{-t/2}\le t^v\,2^{-t/2}\,,$$

which gives (4.5). If $v>t/2$, then (4.5) holds because its left side is at most 1 and $(t/m)^v 2^{-t/2}=(t/m)^{v-t/2}\,(t/(2m))^{t/2}\ge 1$, as $2m\le 1\le t$. □

5 The induction over scales

Scales. Let $\alpha_0\ge 40$ and, for $n\ge 0$,

$$M_n := \alpha_n^\kappa, \qquad \alpha_{n+1} := M_n\alpha_n=\alpha_n^5, \qquad D_n := \frac{7M_n\alpha_n}{a-1},$$
$$\eta_n := \frac{335\alpha_n^2}{D_n}\le 48a\,\alpha_n^{1-\kappa}, \qquad q_n := \frac{129\mu\alpha_n^2}{D_n}\le 19a\mu\,\alpha_n^{1-\kappa}, \qquad t_n := \lceil 2\gamma_0\log_2\alpha_n\rceil,$$
$$\eta_n' := \frac{\eta_0'}{(n+1)^2}, \qquad \eta_0' := \min\Big\{\frac14,\frac{\pi}{4\mu(a-1)}\Big\}, \qquad m_n := \frac{2\mu\eta_n'}{\pi},$$

and $r_0:=2$, $r_{n+1}:=r_n-\eta_n-\eta_n'$. Then $(a-1)m_n\le 1/2$ and $2^{-t_n/2}\le\alpha_n^{-\gamma_0}$. We assume that α_0 is so large that for all $n\ge 0$

$$M_n\ge\max\{100,6a\}\,,\qquad \sum_{n\ge0}\eta_n\le 0.1\,, \tag{5.1}$$

$$a^2\,q_n\,\frac{t_n^2}{m_n^2}\,\alpha_n^{\bar\sigma(a-2)}\le 1\,, \tag{5.2}$$

$$2^a\,\mathrm{e}\,\big(3\,(104+163\sqrt{a}\,k_c)\big)^a\,\alpha_n^{-\xi}\le 1\,. \tag{5.3}$$

This is possible. For (5.2), the left side is at most $C(\mu)\,(n+1)^4(\log\alpha_n)^2\alpha_n^{-0.2}$ by (1.4), and $n+1\le 1+\log\log\alpha_n$, so it is at most 1 as soon as α_n exceeds a constant depending on μ, hence for all n if α_0 does. The others are immediate. Since $\sum_n\eta_n'\le\eta_0'\pi^2/6\le 0.42$, we have $r_n\in[1.48,2]$ for all n.

Theorem 5.1 (Induction)**.** *Let α_0 be as above and let $j\in\mathbb{Z}$ be such that $F_j^2(s;\alpha_0)\le\alpha_0^{-\gamma(s)}$ for all $s\ge 0$. Then for all $n\ge 0$ and $s\ge 0$,*

$$F_j^{r_n}(s;\alpha_n)\le\alpha_n^{-\gamma(s)}\,. \tag{5.4}$$

The proof uses two lemmas. In both, n is fixed, (5.4) is assumed for this n, and we write $\alpha = \alpha_n$, $M = M_n$, and use the notation of Section 3.

Lemma 5.2 (Shared lines are free)**.** *For every $x \in \mathbb{R}^2$, every tuple T and all finite sets $\Lambda_1, \dots, \Lambda_a$ of lines with $s_i = |\Lambda_i|$,*

$$\mathbb{P}\Big(\bigcap_{i=1}^{a} G_j^{r_{n+1},\Lambda_i}(x_i,\alpha)\Big) \le 2^a \,\mathrm{e} \prod_{i=1}^{a} \alpha^{-\gamma(s_i)}\,.$$

Proof. We apply Proposition 4.1 with $D = D_n$, $r = r_{n+1}$, $\eta = \eta_n$, $\eta' = \eta'_n$, $t = t_n$, so that $r'' = r_n$ and $\Theta(k) \le \alpha^{-\gamma(k)}$ by (5.4). Since $2^{-t/2} \le \alpha^{-\gamma_0} \le \alpha^{-\gamma(k)}$ and by Lemma 1.2,

$$\Theta\big(s_i + b_i(h)\big) + 2^{-t/2} \le 2\,\alpha^{-\gamma(s_i+b_i(h))} \le 2\,\alpha^{-\gamma(s_i)}\,\alpha^{\bar\sigma b_i(h)}\,.$$

Moreover $\sum_i b_i(h) = \sum_{k<l}(a-l)h_{kl} \le (a-2)\sum_{k<l} h_{kl}$. Hence the right side of (4.1) is at most

$$2^a \prod_i \alpha^{-\gamma(s_i)} \sum_h \prod_{k<l} \frac{\big(q_n\, t_n^2\, m_n^{-2}\, \alpha^{\bar\sigma(a-2)}\big)^{h_{kl}}}{h_{kl}!} = 2^a \prod_i \alpha^{-\gamma(s_i)} \exp\Big\{ \binom{a}{2} q_n \,\frac{t_n^2}{m_n^2}\, \alpha^{\bar\sigma(a-2)} \Big\}\,,$$

and the exponential is at most e by (5.2). □

Lemma 5.3 (Counting)**.** *Let $0 \le s \le k_c$, $x \in \mathbb{R}^2$ and let Λ be a set of at most s lines. For $k \in \bigcup_i K_i$ let $s(k)$ be the number of lines of Λ meeting $B(x + \alpha k, 9\alpha)$. Then*

$$\prod_{i=1}^{a} \sum_{k\in K_i} \alpha^{-\gamma(s(k))} \le \big(3\,(104 + 163\sqrt{a}\,k_c)\big)^a\, \alpha^{-\xi}\, (M\alpha)^{-\gamma(s)}\,.$$

Proof. Let τ_i be the number of lines of Λ that are near-tangent to the ith circle; then $\sum_i \tau_i \le s$. By Lemma 3.2, at most sc_0 points $k \in K_i$ have a region met by a line of Λ that is not near-tangent to the ith circle, and for these $s(k) \le s$. At most $\tau_i c_1 \sqrt{M}$ further points have a region met by some line of Λ, and for these $s(k) \le \tau_i$. For all other points $s(k) = 0$. Since γ is non-increasing,

$$\sum_{k\in K_i} \alpha^{-\gamma(s(k))} \le \underbrace{104M\,\alpha^{-\gamma_0}}_{\text{(A)}} + \underbrace{80\,\tau_i\sqrt{M}\,\alpha^{-\gamma(\tau_i)}}_{\text{(B)}} + \underbrace{163\sqrt{a}\,s\,\alpha^{-\gamma(s)}}_{\text{(C)}}\,.$$

Expanding the product over i gives at most 3^a terms, one for each partition of $\{1,\dots,a\}$ into sets A, B, C on which the summand (A), (B), (C) is chosen; a term vanishes unless $\tau_i \ge 1$ for all $i \in B$ and $s \ge 1$ or $C = \varnothing$. With $M = \alpha^\kappa$, $g := \gamma(s)$ and n_A, n_B, n_C the cardinalities, each term is at most $(104 + 163\sqrt{a}\,k_c)^a\,\alpha^E (M\alpha)^{-g}$ with

$$E := -n_A(\gamma_0 - \kappa) + \sum_{i\in B}\big(\tfrac{\kappa}{2} - \gamma(\tau_i)\big) - n_C g + (1+\kappa)g\,,$$

and it suffices to show $E \le -\xi$. Recall $\gamma_* \le g \le \gamma_0$ and $n_A + n_B + n_C = a$.

Case $n_B = 0$. If $n_C \ge \kappa + 2$, then $E \le -(n_C - 1 - \kappa)g \le -\gamma_* \le -\xi$. If $n_C \le \kappa + 1$, then $n_A \ge a - \kappa - 1$ and $E \le -(a-\kappa-1)(\gamma_0 - \kappa) + (1+\kappa)\gamma_0 \le -\xi$ by (1.3).

Case $n_B \ge 1$. Since $1 \le \tau_i \le s \le k_c$ for $i \in B$ and $\sum_{i\in B}\tau_i \le s$, we have $\sum_{i\in B}\gamma(\tau_i) \ge n_B\gamma_0 - \sigma s = (n_B - 1)\gamma_0 + g$, hence

$$E \le -n_A(\gamma_0 - \kappa) - (n_B - 1)(\gamma_0 - \tfrac{\kappa}{2}) + \tfrac{\kappa}{2} + (\kappa - n_C)g\,.$$

If $n_C \le \kappa$, then $n_A + n_B - 1 \ge a - 1 - \kappa$ and $E \le -(a-1-\kappa)(\gamma_0 - \kappa) + \frac{\kappa}{2} + \kappa\gamma_0 \le -\xi$ by (1.3). If $\kappa + 1 \le n_C \le a - 2$, then $n_A + n_B - 1 \ge 1$ and $E \le -(\gamma_0 - \kappa) + \frac{\kappa}{2} \le -\xi$ by (1.3). If $n_C = a - 1$, then $n_A = 0$, $n_B = 1$ and $E \le \frac{\kappa}{2} - (a-1-\kappa)g \le \frac{\kappa}{2} - (a-1-\kappa)\gamma_* \le -\xi$ by (1.4). □

Proof of Theorem 5.1. For $n = 0$ the claim is the assumption, since $r_0 = 2$. Assume (5.4) for n. For $s > k_c$ the claim for $n+1$ is trivial. Let $s \le k_c$, $x \in \mathbb{R}^2$ and $|\Lambda| \le s$. For a tuple T let Λ_i be the set of lines of Λ meeting $R_i = B(x + \alpha k_i, 9\alpha)$, so that $|\Lambda_i| = s(k_i)$ and $G_j^{r_{n+1},\Lambda}(x_i, \alpha) = G_j^{r_{n+1},\Lambda_i}(x_i, \alpha)$. By Lemmas 3.1 and 5.2,

$$\mathbb{P}(G_j^{r_{n+1},\Lambda}(x, \alpha_{n+1})) \le \sum_T 2^a \,\mathrm{e} \prod_{i=1}^{a} \alpha^{-\gamma(s(k_i))} = 2^a \,\mathrm{e} \prod_{i=1}^{a} \sum_{k \in K_i} \alpha^{-\gamma(s(k))} ,$$

which is at most $\alpha_{n+1}^{-\gamma(s)}$ by Lemma 5.3 and (5.3). □

Proof of Theorem 1.1. Let $\alpha_0 = \alpha_0(\mu)$ satisfy the requirements of this section, $j_0 = j_0(\mu, \alpha_0)$ from Lemma 2.3, $j := j_0 + k_c$ and $\lambda_0 := 1/(68(j-1))$. By Lemma 2.3 the assumption of Theorem 5.1 holds, so $F_j^{r_n}(0; \alpha_n) \le \alpha_n^{-\gamma_0} = \alpha_n^{-7}$ for all n. For $\lambda \le \lambda_0$ we have $\sqrt{2}\lambda(j-1) \le \sqrt{2}/68 < 1/48 = p$, and Lemma 2.2 bounds the probability in the theorem by $F_j^1(0; \alpha_n) \le F_j^{r_n}(0; \alpha_n)$. □

6 Where an error would most likely be

The argument is short for the statement it claims, and earlier schemes of ours that looked correct at this stage failed in the details. We list the steps that carry the weight, in the order in which we would check them.

1. *Step 3 of Proposition 4.1.* The bound (4.4) places the supremum over the free lines outside the average over the offsets of the straightened lines. This rests on the ordering of the classes by their two smallest region indices and on integrating the regions in decreasing order. If a variable $u_i(\ell)$ entered the factor of a region that has not yet been integrated, other than region i, the bound would fail.

2. *Step 4 of Proposition 4.1.* An additional line with uniform offset and fixed direction is compared, by the Mecke formula, with the lines of $\mathcal{L}$ in a direction window of width 2ψ around this direction. The price is the factor t/m per line and the enlargement of the radius by η'. The radii are summable because $\eta'_n \propto (n+1)^{-2}$, and the resulting factor $(n+1)^4$ is absorbed in (5.2).

3. *Lemma 3.3.* The product structure of the measure of the lines meeting two regions in the coordinates (u_1, u_2), and the fact that within a region such a line deviates from the line of direction e with the same offset by at most $\eta = 335\alpha^2/D$. This is where $\kappa > 1$ is used: η_n is of order $\alpha_n^{1-\kappa}$.

4. *The hypothesis with additional lines.* The induction hypothesis (5.4) is a bound uniform in the positions of s additional lines, with the exponent $\gamma(s) = \gamma_0 - \sigma s$ for $s \le k_c$ and the trivial bound beyond. It holds at the base scale by (2.1) because the level j is chosen k_c above j_0. In Lemma 5.2 a shared line has weight $q_n t_n^2/m_n^2$, of order $\alpha^{1-\kappa}$ up to logarithms, and worsens the product of the bounds by at most $\alpha^{\bar\sigma(a-2)}$; this is the inequality $\bar\sigma(a-2) < \kappa - 1$ in (1.4). In Lemma 5.3 the lines of Λ are distributed over the sub-regions; the case distinction there should be rechecked, in particular the role of near-tangent lines and the use of $\sum_i \tau_i \le s$. A numerical evaluation of the exponent E over all partitions and the extreme choices of the τ_i gave a maximum of -0.2.

5. *Consistency checks we made.* For a line process with a single direction the model percolates at every level, and the argument fails as it should, because Lemma 3.3 (i) uses the density

of ν in the direction. The bounds are consistent with the lower bound for a crossing along a bundle of j nearly parallel lines, which is of order α^{2-j} with $j \geq k_c + j_0$, far below α^{-7}.

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Use of AI tools

The proofs, computations and the text of this note were developed with substantial assistance of the AI systems Claude (Anthropic) and ChatGPT Astra 6 (OpenAI), in an interactive process directed by the authors. The argument has not been verified independently of these tools.

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